1)

```
punit@PUNITHNIKHILGV:~$ cd "/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
_Code_Analysis"
source venv/bin/activate
(venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
_Code_Analysis$ python -m pip show pylint bandit flake8
Name: pylint
Version: 4.1.3
Summary: python code static checker
Home-page:
Author:
Author-email: Python Code Quality Authority <code-quality@python.org>
License:
Location: /mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static_Code_Analysis/ven
v/lib/python3.12/site-packages
Requires: astroid, dill, dill, isort, mccabe, platformdirs, tomlkit
Required-by:
---
Name: bandit
Version: 1.9.4
Summary: Security oriented static analyser for python code.
Home-page: https://bandit.readthedocs.io/
Author: PyCQA
Author-email: code-quality@python.org
License: Apache-2.0
Location: /mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static_Code_Analysis/ven
v/lib/python3.12/site-packages
Requires: PyYAML, rich, stevedore
Required-by:
---
Name: flake8
Version: 7.4.1
Summary: the modular source code checker: pep8 pyflakes and co
Home-page: https://github.com/pycqa/flake8
Author: Tarek Ziade
Author-email: tarek@ziade.org
License: MIT
Location: /mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static_Code_Analysis/ven
v/lib/python3.12/site-packages
Requires: mccabe, pycodestyle, pyflakes
Required-by:
(venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
_Code_Analysis$
```

2)

```
 should be 'parking_slots.get(slot_id) is None' (singleton-comparison)
parking_lot_system_original.py:40:4: R1703: The if statement can be replaced with 'return b
ool(test)' (simplifiable-if-statement)
parking_lot_system_original.py:40:4: R1705: Unnecessary "else" after "return", remove the "
else" and de-indent the code inside it (no-else-return)
parking_lot_system_original.py:37:4: W0612: Unused variable 'overtime_fee' (unused-variable
)
parking_lot_system_original.py:46:0: C0116: Missing function or method docstring (missing-f
unction-docstring)
parking_lot_system_original.py:46:0: C0103: Function name "calculateParkingFee" doesn't con
form to snake_case naming style (invalid-name)
parking_lot_system_original.py:48:16: W0123: Use of eval (eval-used)
parking_lot_system_original.py:47:4: W0612: Unused variable 'base_rate' (unused-variable)
parking_lot_system_original.py:52:0: C0116: Missing function or method docstring (missing-f
unction-docstring)
parking_lot_system_original.py:52:0: C0103: Function name "generateVehicleHash" doesn't con
form to snake_case naming style (invalid-name)
parking_lot_system_original.py:56:0: C0116: Missing function or method docstring (missing-f
unction-docstring)
parking_lot_system_original.py:56:0: C0103: Function name "generateTicketPin" doesn't confo
rm to snake_case naming style (invalid-name)
parking_lot_system_original.py:60:0: C0116: Missing function or method docstring (missing-f
unction-docstring)
parking_lot_system_original.py:60:0: C0103: Function name "exportParkingManifest" doesn't c
onform to snake_case naming style (invalid-name)
parking_lot_system_original.py:61:8: W1514: Using open without explicitly specifying an enc
oding (unspecified-encoding)
parking_lot_system_original.py:61:8: R1732: Consider using 'with' for resource-allocating o
perations (consider-using-with)
parking_lot_system_original.py:67:0: C0116: Missing function or method docstring (missing-f
unction-docstring)
parking_lot_system_original.py:67:0: C0103: Function name "getAvailableSummary" doesn't con
form to snake_case naming style (invalid-name)
parking_lot_system_original.py:67:24: W0621: Redefining name 'parking_slots' from outer sco
pe (line 6) (redefined-outer-name)
parking_lot_system_original.py:69:8: W0612: Unused variable 's' (unused-variable)
parking_lot_system_original.py:75:0: C0116: Missing function or method docstring (missing-f
unction-docstring)
parking_lot_system_original.py:4:0: W0611: Unused import math (unused-import)

---------------------------------
Your code has been rated at 4.26/10

(venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
_Code_Analysis$
```

3)

```
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/
cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-r
andom
   Location: ./parking_lot_system_original.py:57:11
56      def generateTicketPin():
57          return random.randint(1000, 9999)
58

--------------------------------------------------
>> Issue: [B605:start_process_with_a_shell] Starting a process with a shell, possible injec
tion detected, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shel
l.html
   Location: ./parking_lot_system_original.py:64:4
63          f.close()
64          os.system("echo Parking manifest exported to: " + filename)
65

--------------------------------------------------

Code scanned:
        Total lines of code: 66
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXX
X): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 2
                Medium: 1
                High: 2
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 5
Files skipped (0):
(venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
_Code_Analysis$
```

4)

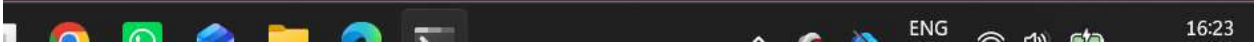
```
                Medium: 0
                High: 5
Files skipped (0):
(venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
_Code_Analysis$ cat flake8_before.txt
parking_lot_system_original.py:4:1: F401 'math' imported but unused
parking_lot_system_original.py:7:80: E501 line too long (82 > 79 characters)
parking_lot_system_original.py:8:80: E501 line too long (83 > 79 characters)
parking_lot_system_original.py:9:80: E501 line too long (82 > 79 characters)
parking_lot_system_original.py:23:80: E501 line too long (131 > 79 characters)
parking_lot_system_original.py:32:5: E722 do not use bare 'except'
parking_lot_system_original.py:37:5: F841 local variable 'overtime_fee' is assigned to but
never used
parking_lot_system_original.py:38:35: E711 comparison to None should be 'if cond is None:'
parking_lot_system_original.py:47:5: F841 local variable 'base_rate' is assigned to but nev
er used
(venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
_Code_Analysis$
```

5)

```
parking_lot_system_original.py:47:5: F841 local variable 'base_rate' is assigned to but nev
er used
(venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
_Code_Analysis$ tail -n 10 pylint_after.txt
************* Module parking_lot_system
parking_lot_system.py:28:0: C0103: Constant name "total_collections" doesn't conform to UPP
ER_CASE naming style (invalid-name)
parking_lot_system.py:33:4: W0603: Using the global statement (global-statement)

-------------------------------------------------------------------
Your code has been rated at 9.66/10 (previous run: 8.23/10, +1.43)

(venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
_Code_Analysis$
```



6)

```
(venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
_Code_Analysis$ cat bandit_after.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.12.3
Run started:2026-10-16 10:45:48.625752+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 93
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXX
X): 0

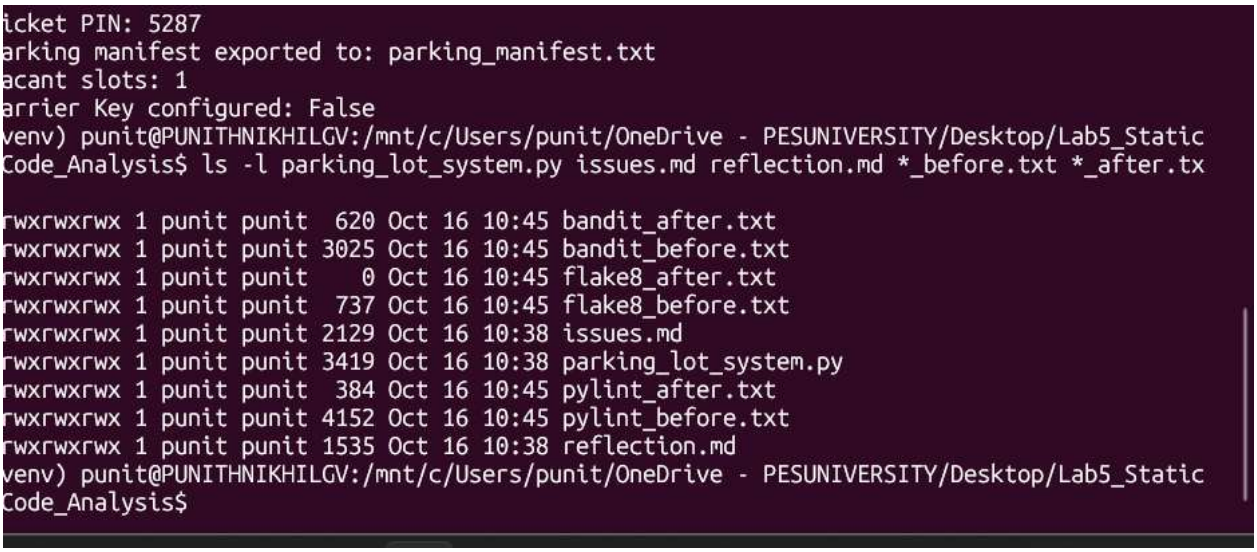
Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
(venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
_Code_Analysis$
```

7)

```
(venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
_Code_Analysis$ python parking_lot_system.py
Is P02 available: False
Surge Parking Fee: 45.0
Vehicle Hash: bd2cd41fd9d9c0045716b50de3536056cd9f8565f9cdb6d1745c639602f3f482
Ticket PIN: 5287
Parking manifest exported to: parking_manifest.txt
Vacant slots: 1
Barrier Key configured: False
(venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
_Code_Analysis$
```



8)



```
icket PIN: 5287
arking manifest exported to: parking_manifest.txt
acant slots: 1
arrier Key configured: False
venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
Code_Analysis$ ls -l parking_lot_system.py issues.md reflection.md *_before.txt *_after.tx

rwxrwxrwx 1 punit punit  620 Oct 16 10:45 bandit_after.txt
rwxrwxrwx 1 punit punit 3025 Oct 16 10:45 bandit_before.txt
rwxrwxrwx 1 punit punit    0 Oct 16 10:45 flake8_after.txt
rwxrwxrwx 1 punit punit  737 Oct 16 10:45 flake8_before.txt
rwxrwxrwx 1 punit punit 2129 Oct 16 10:38 issues.md
rwxrwxrwx 1 punit punit 3419 Oct 16 10:38 parking_lot_system.py
rwxrwxrwx 1 punit punit  384 Oct 16 10:45 pylint_after.txt
rwxrwxrwx 1 punit punit 4152 Oct 16 10:45 pylint_before.txt
rwxrwxrwx 1 punit punit 1535 Oct 16 10:38 reflection.md
venv) punit@PUNITHNIKHILGV:/mnt/c/Users/punit/OneDrive - PESUNIVERSITY/Desktop/Lab5_Static
Code_Analysis$
```